# Attack Monitoring and Blocking: From Transaction Frontrunning to Secure Sequencer

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The results presented in these slides are a team effort and could not have been achieved without the contributions of my students at Zhejiang University and colleagues at BlockSec.

# About Me

- Co-founder of BlockSec and Professor of Zhejiang University
- Research interests
  - DeFi security, Blockchain system security
- Publish: 60+ papers with 9,000+ citations
- Hack and build systems
- Read more: https://yajin.org

# Security Matters in Web3

Despite the bull and bear cycles in the crypto market,
losses caused by exploits and scams have been growing at a rapid pace.

# Security Matters in Web3

Explorer > Security Incidents

https://app.blocksec.com/explorer/security-incidents

## Security Incidents

The attack incidents causing losses exceeding $100K will be documented.

Search For Project Name or Tx Hash of Attack Tx

All | Bookmark

| Project | Loss | Chain | Vulnerability | Date | | |
|---|---|---|---|---|---|---|
| Ronin Bridge | ~ $11.8 M | | Misconfiguration | 2024/08/06 | Root Cause | Share |
| TokenStake (Unknown) | ~ $578 K | | Vulnerable Price Dependency | 2024/08/05 | Root Cause | Share |
| Convergence | ~ $210 K | | Unverified User Input | 2024/08/01 | Root Cause | Share |
| Spectra | ~ $550 K | | Arbitrary Call | 2024/07/23 | Root Cause | Share |
| DeltaPrime | ~ $1 M | | Misconfiguration | 2024/07/22 | Root Cause | Share |
| UPS | ~ $521 K | | Business Logic Flaw | 2024/07/21 | Root Cause | Share |
| WazirX | ~ $207 M | | Compromised Private Key | 2024/07/18 | Root Cause | Share |
| LI.FI | ~ $11 M | | Arbitrary Call | 2024/07/16 | Root Cause | Share |
| Minterest Finance | ~ $1.5 M | | Reentrancy | 2024/07/14 | Root Cause | Share |

BlockSec DeFi Security Incidents Dashboard

# Why Security Incidents are Prevalent

- Economical incentive
  - Hackers can get “paid”. Think about a house full of gold but without a good security system
- Less security-qualified developers
  - Developers are not trained well in security concepts
- DeFi composability: creates more attack vectors

# Why Security Incidents are Prevalent

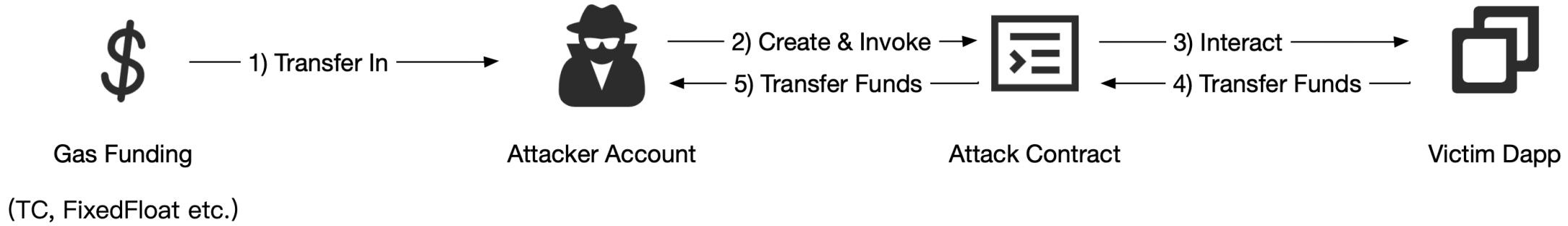
- Openness: everyone can see the code on the chain, and everyone can issue an attack tx if a vulnerability exists
- Anonymity: it’s hard to trace (not impossible) the attacker if he/she is smart enough to hide
- Flashloan: enlarge the money that an attacker can use

# Existing Approach

- Pre-launch
  - Code auditing
  - Fuzzing testing
  - Formal verification
- Post-launch
  - We think there should be an effective approach to detect and **block** hacks after the protocol is deployed

# Steps of a DeFi Attack

- Prepare the attack
- **Launch the attack**
- Launder the profit

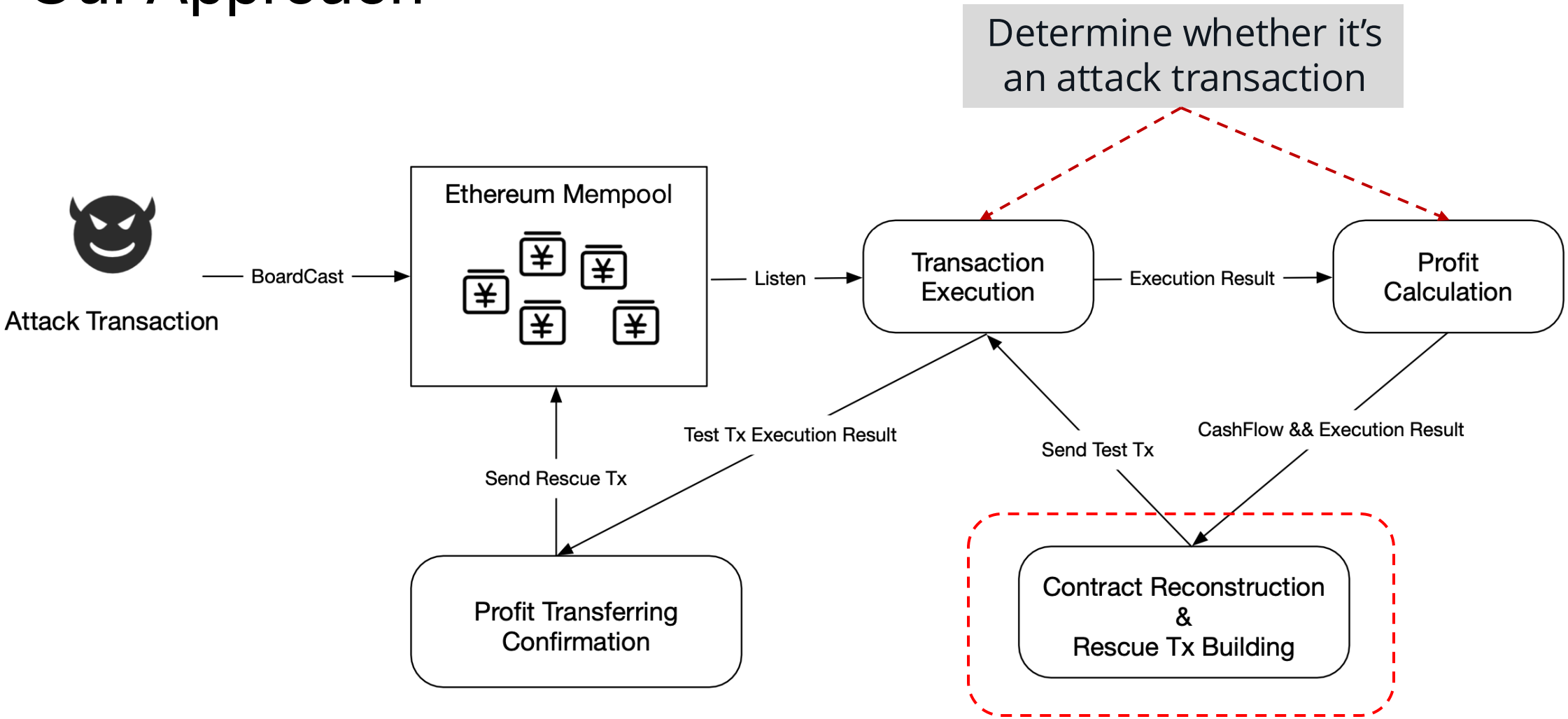


# Our Approach

- The process of a transaction is confirmed
  - T1: A transaction is broadcasted to the p2p network
  - T2: Every node on the network can listen to pending transactions
  - T3: Validators confirm the transaction

Can we **detect and block** the attack transaction during the time window of T1 and T3 (typically 12 seconds in Ethereum)

# Our Approach



# Challenges and Solutions

- How to extract attack logic from the attack contract
  - The attack contract contains verification logic (of the caller) – in a simple way or in an obfuscated way
  - Callback functions

# Challenges and Solutions

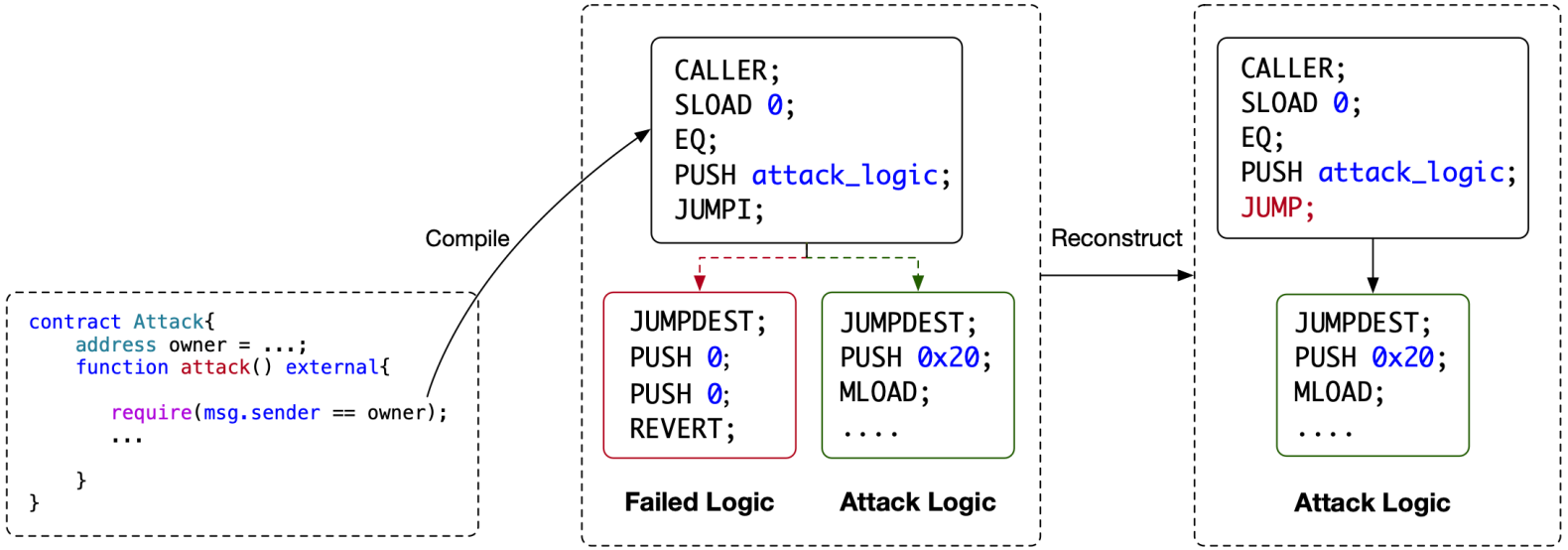
- How to locate and replace revenue addresses
  - Revenue addresses are the ones that get profit during the attack
  - We need to replace them with our ones

# Challenges and Solutions

- How to identify the pre-conditions of an attack
  - Auxiliary contracts
  - Multiple transactions to launch an attack

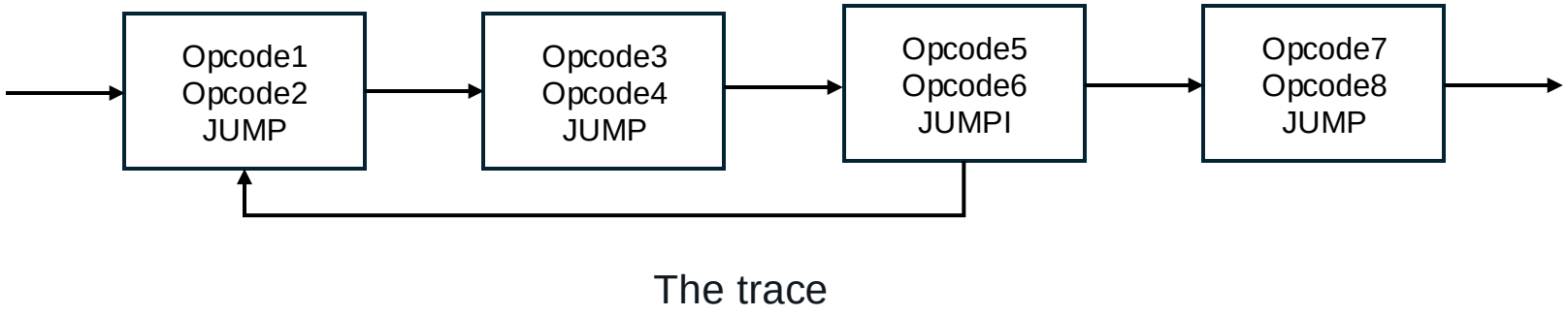
# Task One: Extract and Construct Attack Logic

- Automatically reconstruct a smart contract with the attack logic
  - Attacks usually use some logic to protect their attack contacts



# Task One: Extract and Construct Attack Logic

- Option one: connect the opcode trace of an attack transaction
  - Input: opcode trace of an attack transaction
  - Output: a new smart contract with the trace




Generated contract

Opcode1
Opcode2
Opcode3
Opcode4
Opcode5
Opcode6
Opcode1
Opcode2
Opcode3
Opcode4
Opcode5
Opcode6
Opcode7
Opcode8

# Task One: Extract and Construct Attack Logic

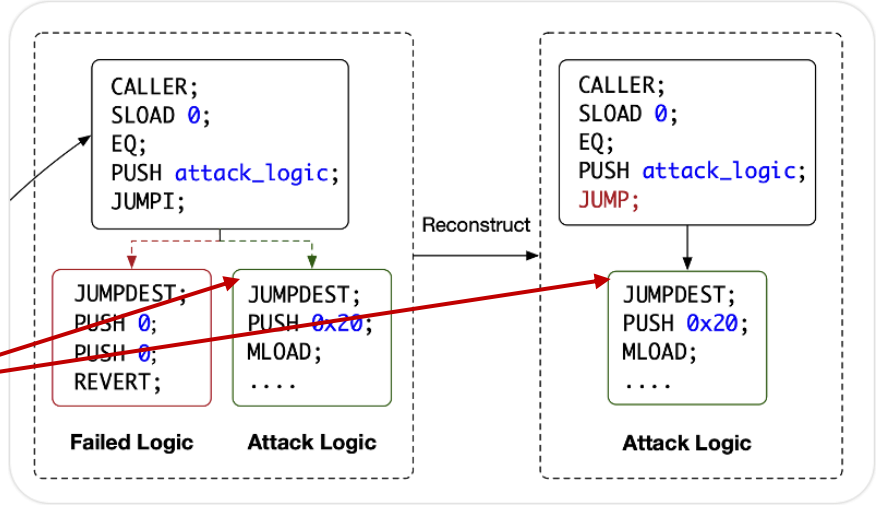
- Option one: connect the opcode trace of an attack transaction
  - It works theoretically but does not work in practice
  - Contract size is limited – it will generate a large contract if the attack has a loop

Opcode1
Opcode2
Opcode3
Opcode4
Opcode5
Opcode6
Opcode1
Opcode2
Opcode3
Opcode4
Opcode5
Opcode6
Opcode7
Opcode8

# Task One: Extract and Construct Attack Logic

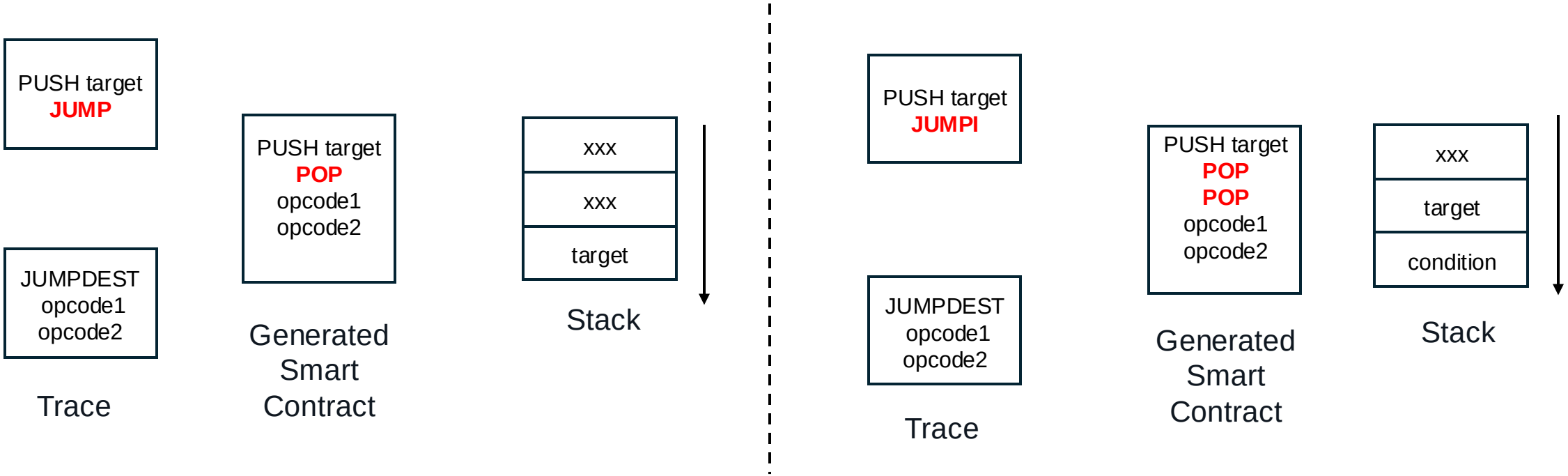
- Option two: reuse the basic block as much as possible
  - It works theoretically and in practice
  - But the process is more complicated than the previous one
    - We need to handle the JUMP instruction and fixup the offsets in the trace and our generated contract

Offsets are different



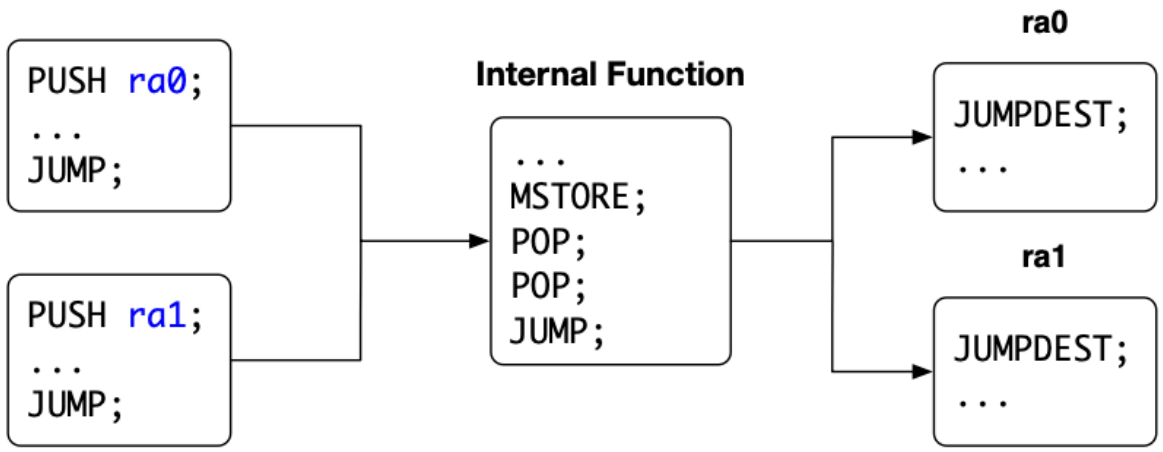
# Freezing Conditional JUMP

- For simple JUMP and JUMPI -> replace with POP



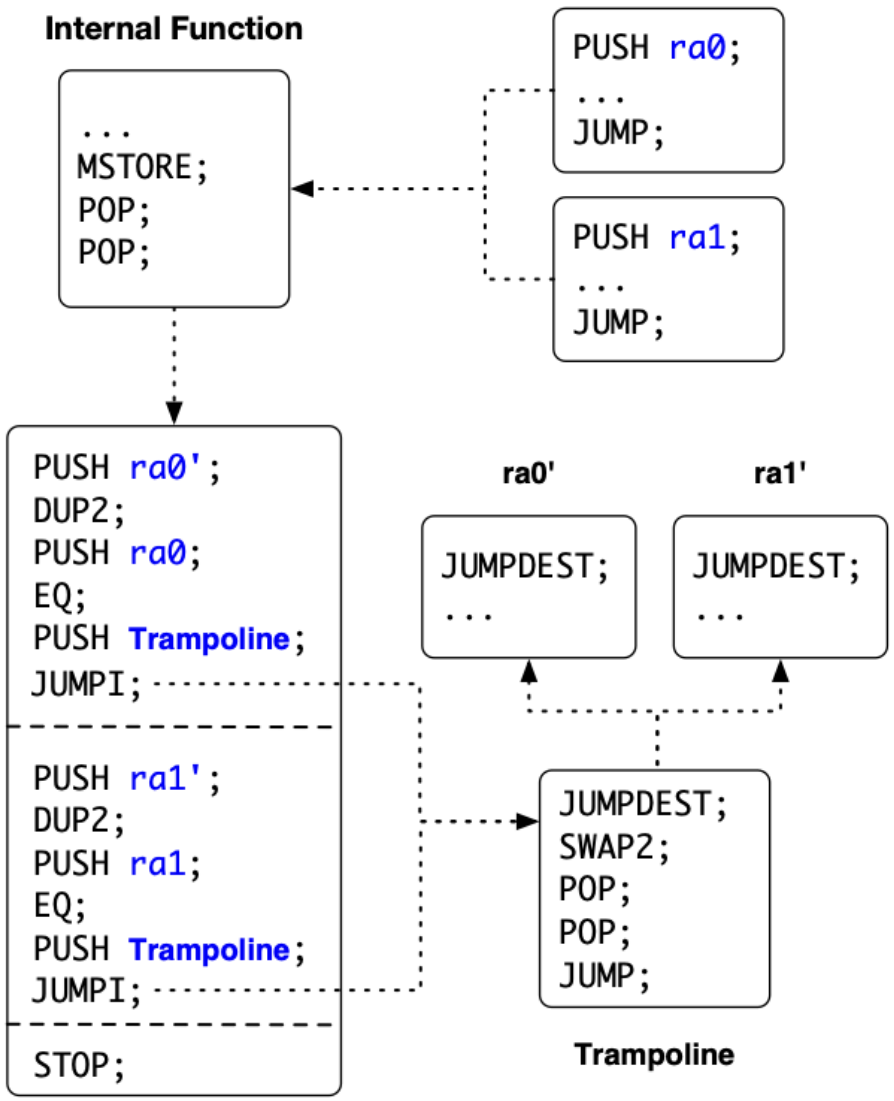
# Freezing Conditional JUMP

- For JUMP with multiple targets
  - The real target is determined in the runtime -> from which basic block



# Freezing Conditional JUMP

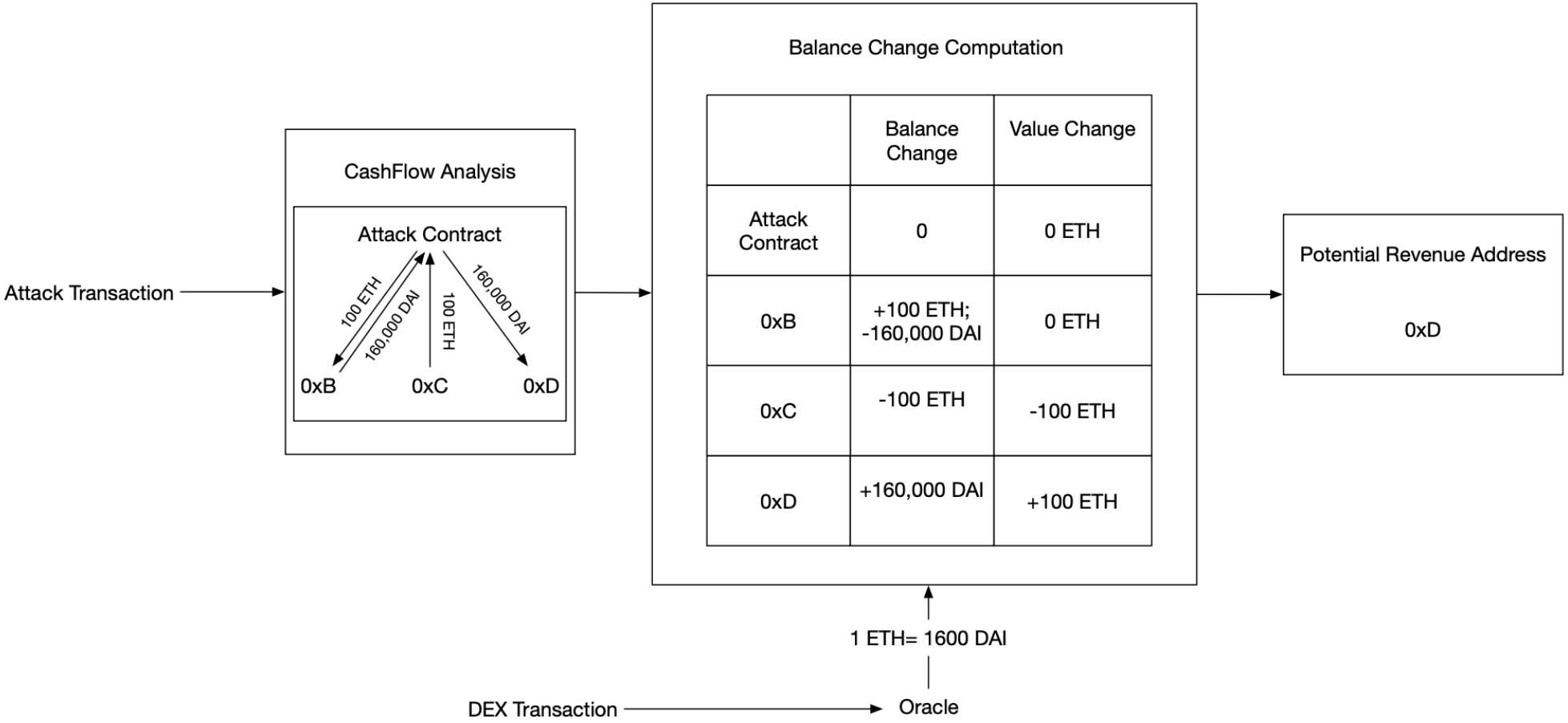
- We compare the basic block and use a trampoline to relocate the destination block
  - **ra0 -> ra0'**
  - **ra1 -> ra1'**
- Also, we need to make the stack balance after the trampoline



# Task Two: Identify and Replace Revenue Addresses

- Revenue addresses
  - The ones that balance changes positively, *and* ***replacing them does not affect the attack logic***
- How to identify them: try-and-catch
  - Calculate the balance changes
  - Replace the ones with positive balance change to see what happens
- When: Replace them when they are pushed on the stack

# Task Two: Identify and Replace Revenue Addresses

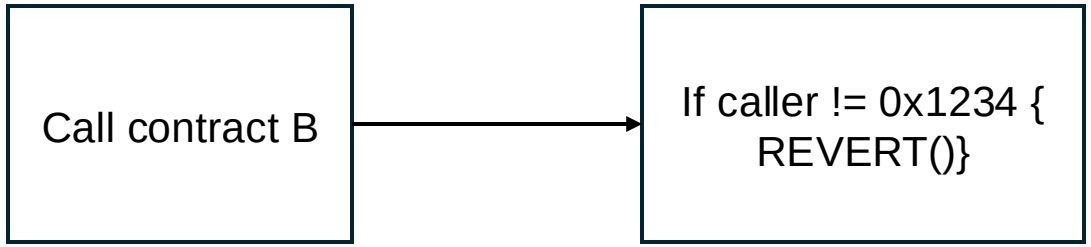


# Task Three: Identify Pre-conditions

- Auxiliary contracts
  - Contracts created by the attacker before or during the attack process – they also contain the attack logic
- Multiple transactions
  - An attack can consist of multiple transactions: T1, T2 prepare the attack and T3 launches the attack

# Identify Auxiliary contracts

- Created during the attack: find CREATE/CREATE2 instruction
- Created before the attack: use REVERT to locate them
  - The auxiliary will revert during the interaction (since we change the caller when invoking the auxiliary contract)
  - We track all the reverted internal calls to identify the auxiliary contract




Auxiliary contract

# Identify Multiple Transactions

- Attackers can split the attack transaction into multiple ones

EGD Finance Attack Incident

# Identify Multiple Transactions

- Our method
  - We retrieve the state dependency search method to locate all preparation transactions
  - State-dependent: a transaction is using a storage changed by a previous transaction
  - Optimizations: cache state change called by a tx in (24 hours)

# Evaluation

- Effectiveness
  - If our system can successfully synthesize the attack tx and contract with replaced revenue addresses, then our system is effective in blocking that hack
- Dataset
  - Historical attacks in DeFiHackLabs: June 2020 to Februray 2023
  - Real attacks

# Evaluation

- Dataset: 117 attacks in total, we use 87 of them
  - The attack crosses different blocks
  - They require some special tokens
- Among 87 attacks, we can successfully generate the attack contracts

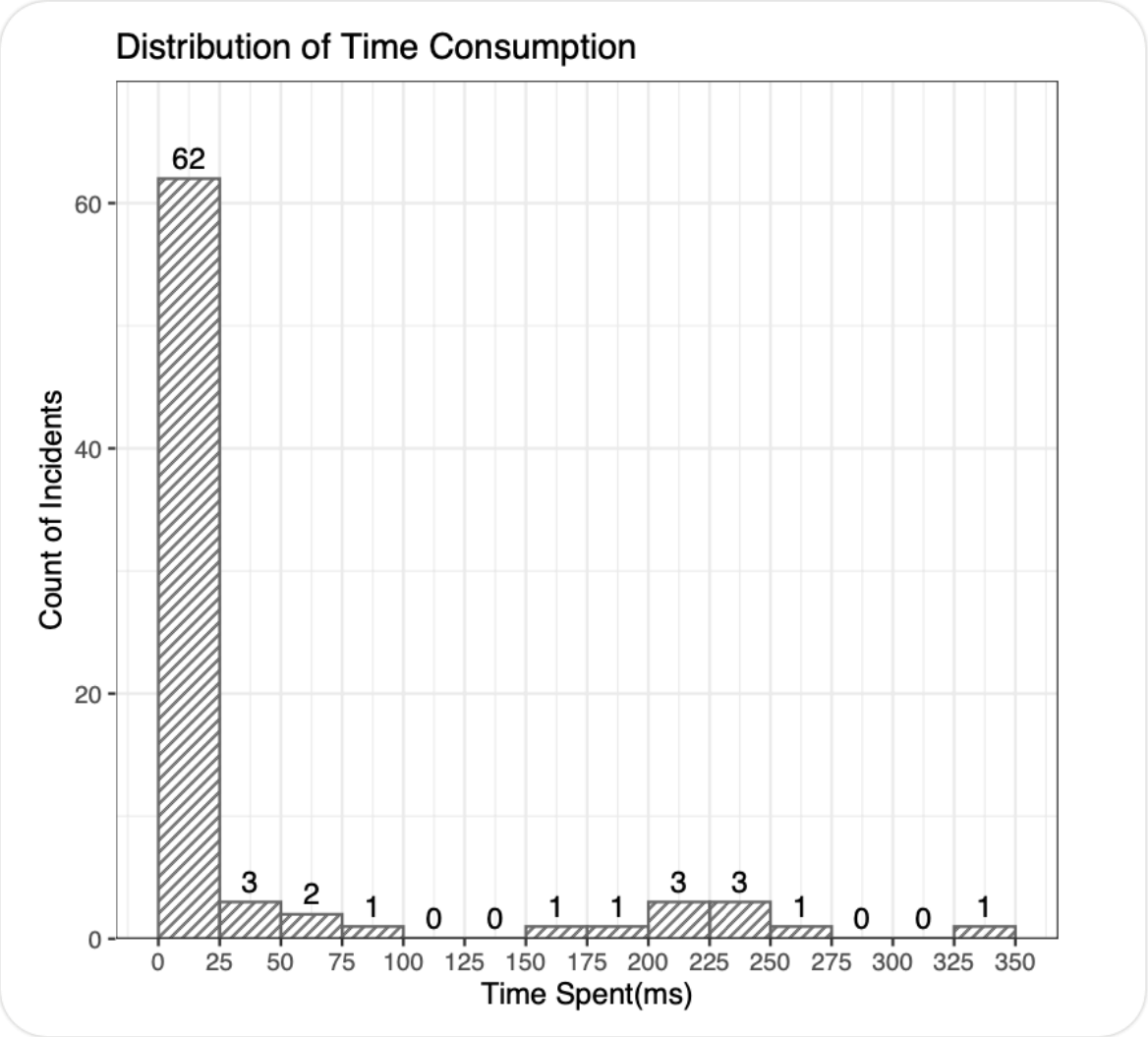
| Feasibility | Category | Success | Total |
|---|---|---|---|
| Not Feasible | Unable to finish in a single block | 0 | 29 |
| | Require special capital | 0 | 4 |
| Feasible | Common attack | 47 | 50 |
| | Involve anti-front-running strategy | 31 | 34 |

# Evaluation

- Reasons for failed cases
  - Attack tx was too complicated with many auxiliary contracts – the generated contract was too big (bigger than the maximum allowed contract size)

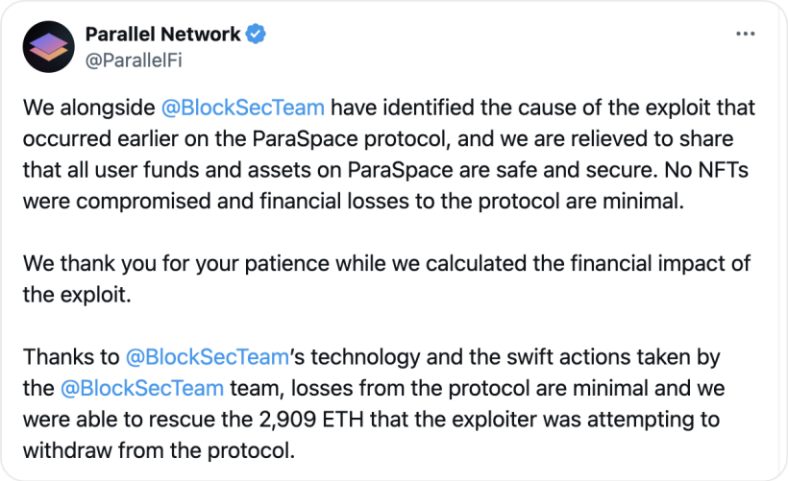
# Evaluation

- Efficiency
  - How quick to reconstruct the attack contract
    - More than 80% cases are finished in less than 25 ms



# Real Cases

- Blocked hacks and rescued more than 20 million USDs
  - Representative ones: ParaSpace: 5 Million, Saddle Finance: 3.8 million




https://blocksec.com/blog/lead-in-phalcon-s-hack-blocking-saga

# From Frontrunning to Phalcon

- The technique is good, but …
  - What about the legal issue
  - What about the competition between our system and other MEV bots
- We need another solution from the perspective of a product

# BlockSec Phalcon

- A platform to monitor and **block** hacks
  - Hack blocking tx can be **triggered automatically** from the protocol
  - The key to the system it to **monitor the hacks with low false positives and low false negatives**

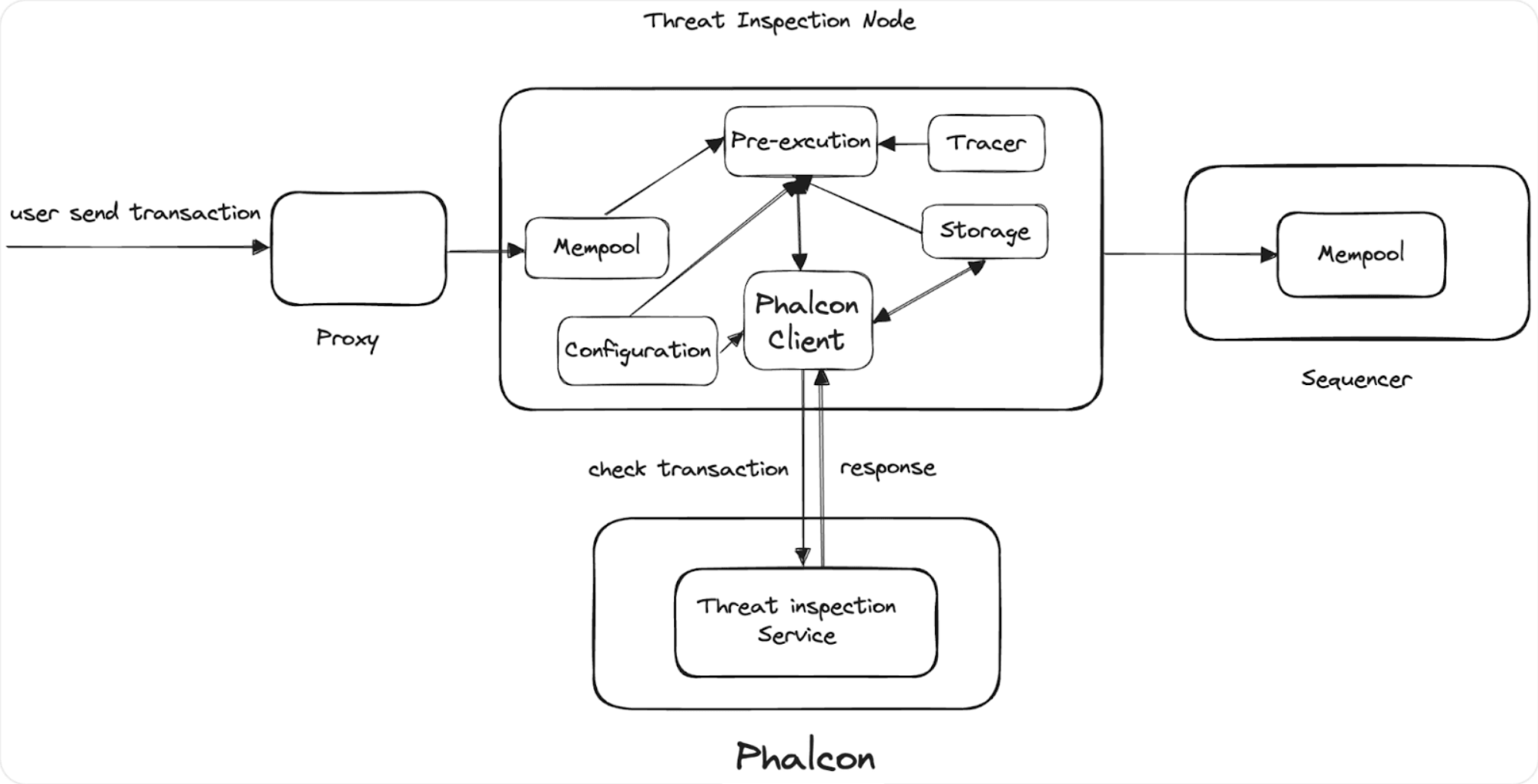
# BlockSec Phalcon

- How to reduce false positives
- How to reduce false negatives

# From Phalcon to STOP (**S**equencer **T**hreat **O**verwatch **P**rogram)

- STOP is the solution to secure L2 chains
- Sequencer plays the role of validator
  - Responsible for packing the transactions inside a block
- We can enable attack detection and blocking inside sequencer

# STOP (**S**equencer **T**hreat **O**verwatch **P**rogram)



# From Phalcon to STOP (**S**equencer **T**hreat **O**verwatch **P**rogram)

- Challenges
  - Attack detection accuracy
  - Latency
  - Transactions from L1 -> L2

# From Phalcon to STOP (**S**equencer **T**hreat **O**verwatch **P**rogram)

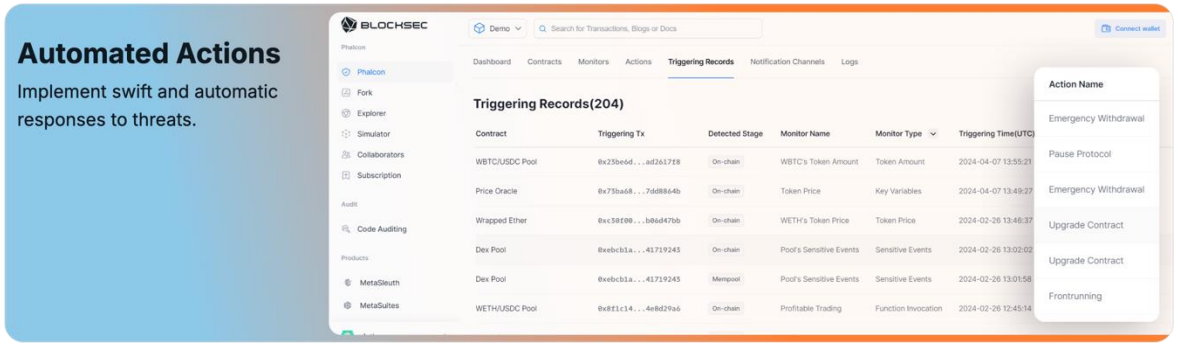
- The first L2 with inborn sequencer-level security

# About BlockSec

- Full stack security providers
- For protocols and chains
  - Pre-launch: code auditing
  - Post-launch
    - Attack monitoring and blocking: Phalcon
    - Crypto tracking and investigation: MetaSleuth (https://metasleuth.io)

# BlockSec Phalcon

- BlockSec Phalcon (https://blocksec.com/phalcon)
  - Gain early access to precise attack intelligence
  - Automated attack blocking with customized actions

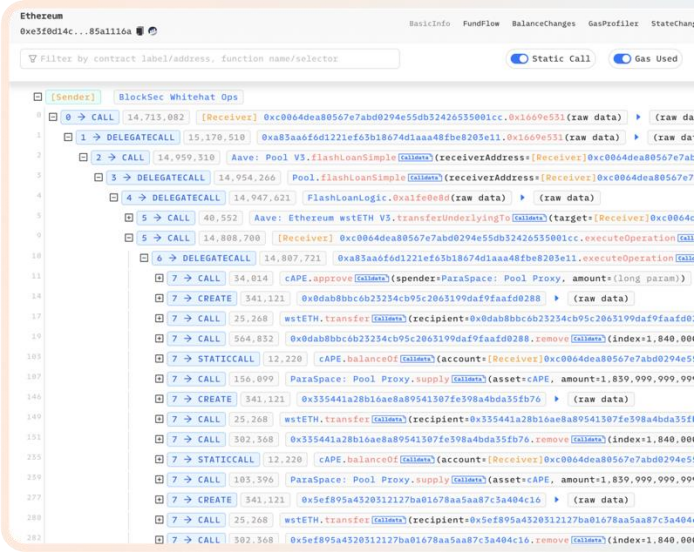


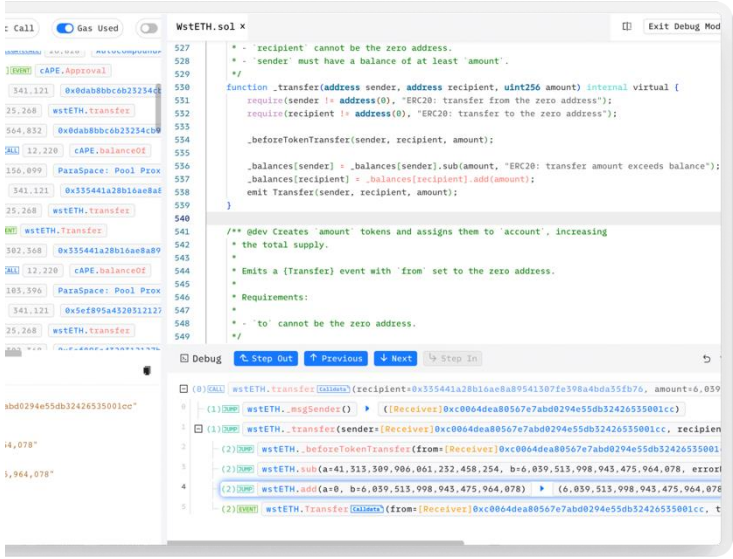
# MetaSleuth

- https://metasleuth.io

# Phalcon Explorer

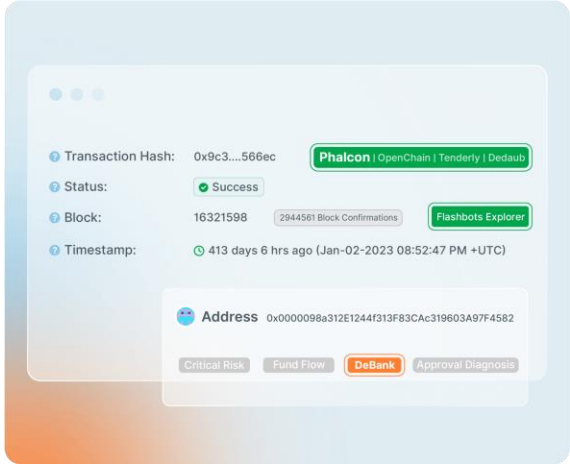
- Free transaction analysis tool for security/MEV researchers – Solana is supported - https://blocksec.com/explorer

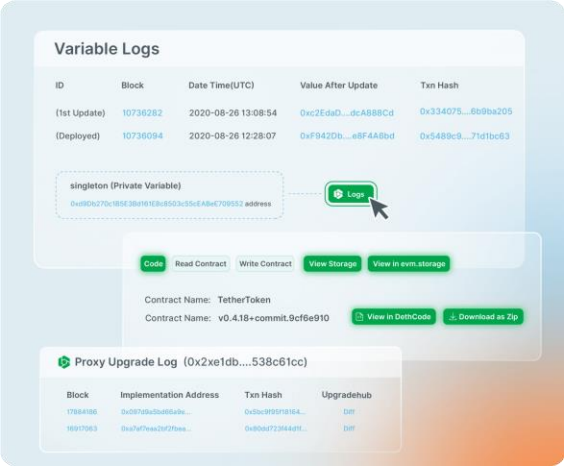




# MetaSuites

- A free extension for builders (https://blocksec.com/metasuites)
  - Address labels
  - Fund flows
  - Aggregator for popular web3 tools
  - Development accelerator
  - It supports **Solana**



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